SAVITRIBAI PHULE PUNE UNIVERSITY

PROJECT STAGE II REPORT ON

"Sign Language to voice Converter"

SUBMITTED TOWARDS THE
PARTIAL FULFILMENT OF THE REQUIREMENTS OF

BACHELOR OF ENGINEERING

(Electronics & Telecommunication Engineering)

BY

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Under The Guidance of

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DEPARTMENT OF ELECTRONICS & TELECOMMUNICATION ENGINEERING

Maratha Vidya Prasarak Samaj's

Karmaveer Adv. Baburao Ganpatrao Thakare
College of Engineering, Nashik-13

Academic Year: 2025-2026

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To educate and train common masses through undergraduate, postgraduate, research programs by inculcating the values for discipline, quality, transparency and foster career and professional development for employment thereby contributing to the development of society.

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5. To provide suitable infrastructure and facilities in tuned with advancing technological evaluation.

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Department of Electronics & Telecommunication Engineering

Certificate

This is to certify that the Project Entitled

"Sign Language to voice Converter"

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Abstract

This project presents the design and development of a Sign Language to Voice Converter using an ESP32 microcontroller to assist individuals with speech and hearing impairments in communicating effectively. The system is designed to translate hand gestures into both text and audible speech in real time. The system employs flex sensors mounted on a wearable glove to detect finger bending patterns corresponding to sign language gestures. In addition, an MPU6050 is integrated to capture hand motion and orientation, improving the accuracy of gesture recognition. The analog signals from the flex sensors are read using the built-in Analog-to-Digital Converter (ADC) of the ESP32, while motion data is acquired via the I2C interface. This project demonstrates a low-cost, compact, and efficient embedded system capable of real-time gesture-to-speech conversion. By combining finger movement and hand motion detection, the system provides improved accuracy and serves as an effective assistive communication tool, helping bridge the gap between speech-impaired individuals and the general population.

Keywords

ESP32, Flex Sensors, MPU6050, DF Player Mini, ADC, Gesture Recognition.

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Abbreviations

SLVC	Sign Language to Voice Converter
PIC	Peripheral Interface Controller
ESP32	Espressif System 32-bit Microcontroller
DFPlayerMini	MP3 Audio Playback Module
LCD	Liquid Crystal Display
ML	Machine Learning
TTS	Text To Speech
ADC	Analog to Digital Converter
VCC	Voltage Common Collector
GND	Ground
AC	Alternating Current
DC	Direct Current

Chapter 1

Introduction

1.1 Necessity:

Communication plays a crucial role in human interaction, but individuals with speech and hearing disabilities often struggle to express their thoughts effectively. Since most people are not familiar with sign language, a significant communication gap exists between speech-impaired individuals and the general population. Therefore, there is a strong need for a system that can automatically convert hand gestures into speech or text. The proposed Sign Language to Voice Converter using the ESP32 aims to fulfil this necessity. By using flex sensors attached to a glove, the system detects finger movements corresponding to different sign gestures. Additionally, a MPU6050 is used to detect hand motion and orientation, improving the accuracy of gesture recognition. These signals are processed by the ESP32, which interprets them and generates the corresponding text on an LCD display and voice output using the DF Player Mini. This enables smooth and effective communication between speech-impaired individuals and others.

1.2 Need of Project:

People with speech and hearing impairments face major challenges when communicating with those unfamiliar with sign language. To assist them, there is a need for a low-cost, portable, and reliable device that can recognize hand

gestures and translate them into meaningful speech or text in real time. The Sign Language to Voice Converter using the ESP32 fulfils this requirement by converting the analog data from flex sensors into digital signals using the built-in ADC (Analog-to-Digital Converter) of the ESP32. In addition, a MPU6050 is used to capture hand movement and orientation, which improves gesture recognition accuracy. These processed signals are then used to generate audible voice output through the DF Player Mini and display the corresponding text on an LCD screen, thereby bridging the communication gap and promoting inclusivity in society.

1.3 Motivation:

Communication is an essential aspect of human life, but individuals with speech and hearing impairments often face significant challenges in expressing their ideas and emotions. They primarily rely on sign language, which is not widely understood by the general population. This communication gap creates barriers in education, healthcare, and daily social interactions. This limitation inspired the development of a Sign Language to Voice Converter using the ESP32, which can interpret hand gestures and convert them into both speech and text. The project aims to design an affordable, reliable, and portable assistive device using flex sensors along with a MPU6050 for improved gesture detection, and voice playback technology using the DF Player Mini. The main motivation behind this work is to utilize embedded system design and modern microcontroller technology to create an innovative solution that enhances social inclusion and enables differently-abled individuals to communicate independently and confidently.

1.4 Problem Statement:

Communication between deaf and mute individuals and others is difficult due to limited understanding of sign language. Existing solutions are often complex or expensive. There is a need for a simple system that can recognize predefined hand gestures and convert them into voice. This project uses sensors and an ESP32 to provide real-time gesture-to-speech conversion.

1.5 Objectives:

- To design a wearable glove integrated with flex sensors to detect finger movements representing predefined sign language gestures.
- To use an ESP32 for acquiring and processing flex sensor data and identifying specific gesture patterns.
- To integrate a motion sensor such as MPU6050 for detecting hand movement and improving gesture recognition accuracy.
- To convert recognized gestures into speech output using a DF Player Mini and speaker for effective audio communication.
- To display the identified gesture text on a 16×2 LCD using the I2C interface for real-time visual feedback.
- To develop a low-cost, reliable, and portable assistive system that bridges the communication gap between sign language users and non-signers.

1.6 Advantages and Limitations:

1.6.1 Advantages:

- **Assistive Communication:** Enables speech- and hearing-impaired individuals to communicate effectively through gesture-based voice conversion.
- **Low Cost and Accessibility:** Built using affordable and easily available components like ESP32, flex sensors, and LCD, making the system economical and easy to implement.
- **Compact and Portable Design:** Lightweight, glove-based system that can be easily carried and used in daily life.
- **Easy Interfacing:** The ESP32 supports multiple interfaces (ADC, I2C, UART), allowing simple and efficient integration with sensors, LCD, and DF Player Mini.
- **Improved Accuracy:** Integration of motion sensing using MPU6050 enhances gesture detection by combining finger movement and hand orientation.
- **Upgradeable System:** The system can be further enhanced using machine learning, additional sensors, or wireless connectivity for advanced gesture recognition.
- **Educational Value:** Demonstrates important embedded system concepts such as ADC conversion, sensor interfacing, I2C communication, and serial communication.

1.6.2 Limitations:

- **Sensor Calibration Required:** Flex sensors must be properly calibrated to ensure accurate gesture detection and to avoid incorrect readings.
- **Limited Gesture Recognition:** The system recognizes only a predefined set of gestures and does not support complete sign language vocabulary or dynamic gesture learning.
- **Display Constraints:** The 16×2 LCD can display only a limited number of characters, restricting the amount of information shown at a time.
- **Pre-recorded Audio Output:** Voice output depends on stored audio files in the DF Player Mini, rather than real-time text-to-speech conversion.
- **User Variation:** Accuracy may vary depending on hand size, finger flexibility, and differences in how users perform gestures.
- **Wired Prototype Design:** The use of wired connections and external modules may reduce comfort and flexibility for long-term wearable use.

1.7 Applications:

1. **Assistive Device for Disabled:** Helps speech- and hearing-impaired individuals communicate effectively through gesture-to-voice translation.
2. **Educational Institutions:** Can be used in special schools to teach and demonstrate gesture-based communication systems.
3. **Healthcare Sector:** Useful for patients with speech loss or neurological disorders to express basic needs.

4. **Human–Machine Interaction:** Can be used in research and automation projects involving gesture-based control.
5. **Rehabilitation Centers:** Assists therapists and patients in communication and recovery sessions.

1.8 Organization of Project Report:

Chapter 1 – Introduction: This chapter provides an overview of the project, including background, need, motivation, problem statement, objectives, advantages, limitations, and applications of the Sign Language to Voice Converter system.

Chapter 2 – Literature Review: This chapter reviews previous research work, existing systems, and technologies related to sign language recognition and gesture-based communication. It also identifies the research gaps that motivated the development of this project.

Chapter 3 – Analysis and Design Approach: This chapter explains the proposed system, its working principle, and overall design methodology. It includes the block diagram, system architecture, circuit design, and software implementation using the ESP32, flex sensors, LCD, and DF Player Mini.

Chapter 4 – Hardware Design: This chapter describes all hardware components used in the system, their specifications, interfacing techniques, and working

principles. It also includes detailed circuit connections and integration of components such as the MPU6050, sensors, display, and audio module.

Chapter 5 – Conclusion and Future Scope: This chapter summarizes the overall work carried out in the project, highlights key achievements and limitations, and suggests future enhancements such as AI-based gesture recognition, mobile application integration, and improved real-time communication features.

Chapter 2

Literature Review

2.1 Smart Glove for Sign Language Recognition Using Flex Sensors and Arduino UNO

Authors: A. Sharma, R.Singh,and P.Deshmukh

Journal: International Journal of Innovative Research in Computer and Communication Engineering (IJIRCCE), 2019

Abstract:

This paper presents the design of a smart glove equipped with flex sensors to recognize hand gestures corresponding to sign language alphabets. The system uses an Arduino UNO microcontroller to process the sensor signals and displays the recognized gesture on an LCD screen. It provides a simple and cost-effective solution for basic sign language interpretation.

Findings:

1. Flex sensors effectively detect finger bending and can be used for accurate gesture recognition.
2. Arduino UNO is sufficient for basic processing tasks but lacks wireless communication features.
3. The system provides reliable real-time feedback through an LCD display.
4. Limited gesture storage capacity restricts the scalability of the design.

5. The study demonstrates feasibility for low-cost assistive technology but lacks IoT integration.

2.2 IoT-Based Sign Language Translation System Using ESP32

Authors: S. Patel, M. Reddy, and A.Ghosh

Journal: International Journal of Advanced Research in Electronics and Communication Engineering (IJARECE), 2021

Abstract:

The paper introduces an IoT-enabled system for converting sign language gestures into text and speech using an ESP32 microcontroller. Flex sensors capture the gesture data, and the ESP32 transmits the recognized text over Wi-Fi to a web dashboard. The recognized gesture is also converted into voice output using an MP3 playback module.

Findings:

1. The ESP32 offers built-in Wi-Fi and Bluetooth, eliminating the need for additional communication modules.
2. The system provides both visual (LCD) and audio feedback for user convenience.
3. IoT integration allows real-time gesture monitoring through a mobile or web interface.
4. Gesture accuracy depends heavily on calibration and sensor stability.
5. Demonstrates how IoT can extend the functionality of traditional assistive devices.

2.3 Real-Time Gesture to Speech Conversion Using Flex Sensors

Authors: R. Joshi, K. Bansal, and D. Mehta

Journal: International Journal of Scientific Research in Engineering and Technology (IJSRET), 2020

Abstract:

This research focuses on the development of a gesture-based communication system using flex sensors and a microcontroller. The system identifies predefined hand gestures and plays corresponding voice messages through a speaker module. It aims to assist speech-impaired individuals by providing real-time gesture-to-speech conversion.

Findings:

1. Flex sensors are efficient for mapping specific gestures when properly calibrated.
2. Real-time speech conversion improves user interaction and accessibility.
3. The design is simple but limited to static gestures.
4. The lack of network connectivity limits data sharing and IoT applications.
5. The study highlights the potential of integrating Wi-Fi-enabled microcontrollers for improved functionality.

2.4 Design and Implementation of Smart Communication Glove Using Flex Sensors and Wi-Fi Module

Authors: N. Kumar, V. Sharma, and T. Pandey

Journal: Journal of Embedded Systems and Robotics (JESR), 2022

Abstract:

This paper presents a Wi-Fi-enabled communication glove that interprets hand gestures and transmits corresponding messages wirelessly. The system uses flex sensors for gesture detection and an ESP8266 Wi-Fi module for data transmission. Recognized gestures are displayed on an LCD and converted into voice output using a speaker.

Findings:

1. The integration of Wi-Fi improves the system's communication range and flexibility.
2. The LCD provides clear and immediate visual feedback for each gesture.
3. The use of multiple sensors allows recognition of complex gestures.
4. The system requires periodic calibration to maintain accuracy.
5. Demonstrates the importance of IoT connectivity for remote accessibility and communication.

2.5 IoT-Based Assistive Glove for Sign to Speech Conversion Using ESP32 and DF Player Mini

Authors: P. Gupta, H. Thakur, and S. Verma

Journal: International Journal of Emerging Technologies and Innovative Research (JETIR), 2023

Abstract:

The authors propose a complete IoT-based sign-to-speech system that utilizes an ESP32 microcontroller, flex sensors, and DF Player Mini for audio output. The device converts detected gestures into corresponding speech and sends recognized text data to a cloud dashboard using Wi-Fi. An LCD display provides instant visual output to the user.

Findings:

1. The ESP32's processing power and wireless features make it ideal for IoT-based gesture translation systems.
2. DF Player Mini enables clear and reliable voice output from pre-recorded files.
3. The system successfully integrates local and cloud-based feedback mechanisms.
4. Flex sensors must be carefully calibrated for consistent performance.
5. The design is practical, scalable, and suitable for low-cost assistive technology applications.

2.6 Review Table:

Sr. No.	Nameof Authors	Journal & Year (Publication Details)	Features / Benefits	Difficulties / Challenges
1	A. Sharma, R. Singh, and P. Deshmukh	International Journal of Innovative Research in Communication Engineering (IJIRCCE), 2019	- Developed a smart glove using flex sensors and Arduino UNO. - Low-cost and easy to implement.	- Limited to basic gestures only. - No wireless or IoT connectivity. - Processing speed restricted by Arduino UNO.
2	S. Patel, M. Reddy, and A. Ghosh	International Journal of Advanced Research in Electronics and Communication Engineering	- Used ESP32 for sign-to-speech conversion. - Provided text and audio output in real time.	- Accuracy depends ons sensor calibration. - Limited number of gestures recognized.
3	R. Joshi, K. Bansal, and D. Mehta	International Journal of Scientific Research in Engineering and Technology (IJSRET), 2020	- Simple design for gesture-to-speech conversion. - Real-time voice feedback using pre-recorded audio files. .	-Recognizes only static gestures. -No visual output display.
4	N. Kumar, V. Sharma, and T. Pandey	Journal of Embedded Systems and Robotics (JESR), 2022	- Wi-Fi-based gesture transmission using ESP8266..	- System complexity increases with more sensors.
5	P. Gupta, H. Thakur, and S. Verma	International Journal of Emerging Technologies and Innovative 2023	- IoT-enabled ESP32-based glove for gesture-to-speech.	-Flexsensors affected by temperature and wear

Table 2.1 Review Table

Chapter 3

Analysis and Design Approach

3.1 System Concept

- The Sign Language to Voice Converter system is designed to assist speech- and hearing-impaired individuals by translating hand gestures into audible speech and text output.
- The glove is embedded with flex sensors that detect finger bending and generate analog signals proportional to the degree of movement.
- These analog signals are fed into the ESP32, which processes them using its built-in Analog-to-Digital Converter (ADC).
- An additional motion sensor, MPU6050, is used to detect hand movement and orientation, improving gesture recognition accuracy.
- The recognized gesture is displayed as text on a 16×2 LCD and simultaneously converted into voice using a DF Player Mini and a speaker.
- This compact, low-cost, and portable system provides an effective communication bridge for differently-abled individuals.

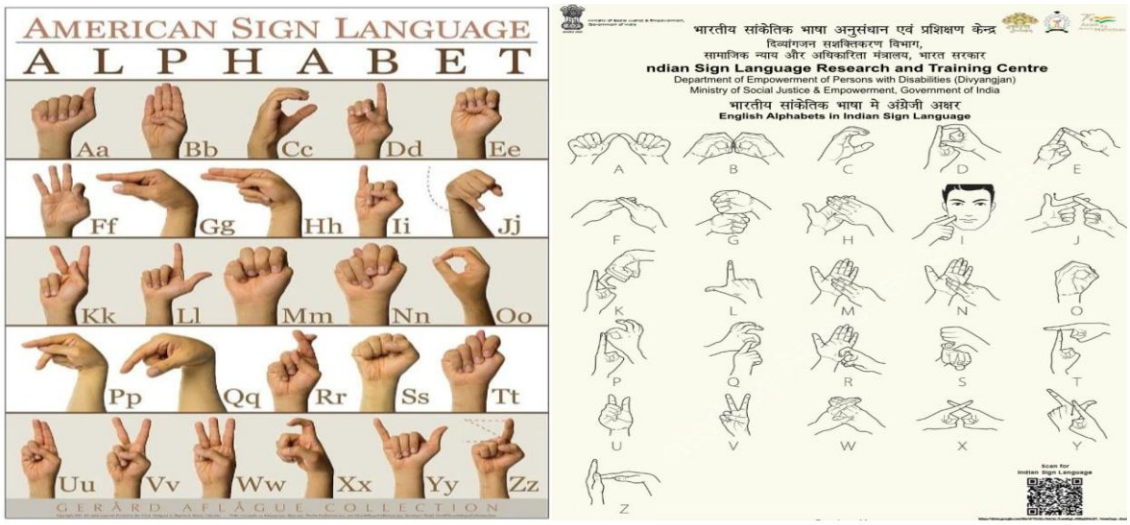

Fig. 3.1 Sign languages

3.2 System Overview:

The system is designed to detect predefined sign language gestures using flex sensors and a motion sensor, process them using the ESP32 microcontroller, and provide corresponding voice and text outputs. This enables effective communication between speech-impaired and normal individuals in real time.

Key Components:

- **Flex Sensors:** Measure finger bending to identify gesture patterns.
- **Microcontroller (ESP32):** Reads analog data from sensors, processes input signals, and controls output devices.
- **MPU6050 Sensor:** Detects hand movement and orientation to enhance gesture recognition.

- **DF Player Mini & Speaker:** Convert recognized gestures into pre-recorded voice messages.
- **16×2 LCD Display:** Displays detected gesture text in real time.
- **Power Supply Unit:** Provides regulated 5V DC for stable system operation.

3.3 System Components:

- Flex Sensors:

Flex sensors act as variable resistors whose resistance changes when bent. Each sensor is attached to a finger and produces a unique voltage pattern for different hand gestures. These analog signals are provided to the ADC pins of the ESP32 for further processing.

- Microcontroller (ESP32):

The ESP32 functions as the central processing unit of the system. It reads analog data from the flex sensors using its built-in ADC, processes the input by comparing it with predefined threshold values, and identifies the corresponding gesture. It then controls output devices such as the LCD and the DF Player Mini.

- MPU6050 Sensor:

The MPU6050 is used to detect hand motion and orientation. It enhances gesture recognition accuracy by providing additional data such as tilt and movement, which helps in identifying dynamic gestures.

- DF Player Mini Module:

This module is used to store and play pre-recorded MP3 voice messages corresponding to recognized gestures. The ESP32 sends serial (UART) commands to the DF Player Mini, and the audio output is played through a speaker.

- 16×2 LCD Display:

The LCD display shows the recognized gesture in text form, providing real-time visual feedback to the user. It is interfaced with the ESP32 using an I2C module, reducing the number of required connection pins.

- Power Supply:

The system is powered by a regulated 5V DC source, such as a USB supply or rechargeable battery. A proper power supply ensures stable operation of the ESP32 and all connected components.

3.4 System Workflow:

1. Gesture Detection and Signal Acquisition:

- The flex sensors embedded in the glove detect finger bending for each gesture.
- Each sensor produces a variable analog voltage proportional to the degree of finger movement.
- These analog signals are transmitted to the ADC pins of the ESP32 for conversion into digital values.
- Additionally, the MPU6050 captures hand orientation and motion data to improve gesture recognition.

2. Data Processing and Gesture Recognition:

- The ESP32 continuously reads digital values converted from the analog sensor inputs.
- These readings are compared with predefined threshold values stored in the program.
- When the input pattern matches a stored gesture, the ESP32 identifies the corresponding gesture accurately using both flex sensor and motion data.

3. Output Generation:

- The recognized gesture is displayed on a 16×2 LCD in real time for visual feedback.
- The DF Player Mini receives UART commands from the ESP32 to play a pre-recorded voice message through a speaker.
- Since the ESP32 has built-in Wi-Fi and Bluetooth, the system can optionally transmit gesture data for IoT applications such as mobile apps or remote monitoring (no need for external modules like ESP8266).

4. Calibration, System Implementation, and Testing:

- Calibration: Each flex sensor is calibrated to map specific voltage ranges to corresponding gestures, ensuring accurate and consistent performance.
- Testing: The system is tested with multiple users performing predefined gestures. Key parameters such as accuracy, response time, and audio clarity are evaluated.

3.5 Project System

1. Selection of Gesture Detection Sensor:

Flex sensors are mounted on each finger of a glove to detect finger bending. The resistance of each sensor varies with the degree of bend—low resistance when straight and high resistance when bent. These analog signals are read using the ADC pins of the ESP32 to identify predefined hand gestures.

2. Microcontroller Integration:

The ESP32 microcontroller is selected due to its high processing speed, multiple ADC channels, and support for various communication protocols. It continuously reads analog values from the flex sensors and compares them with predefined threshold ranges stored in memory. When a gesture pattern matches, the ESP32 triggers corresponding outputs.

3. Motion Detection :

To improve accuracy, a motion sensor such as the MPU6050 can be integrated. It detects hand movement and orientation, allowing the system to recognize dynamic gestures in addition to finger bending.

4. Voice Output Conversion:

A DF Player Mini is interfaced with the ESP32 using UART communication. Pre-recorded audio files are stored on a microSD card. When a gesture is recognized, the ESP32 sends a command to the DF Player Mini to play the corresponding audio through a speaker, enabling real-time voice output.

5. Text Display Interface:

A 16×2 LCD display is used to show the recognized gesture text. It is connected to the ESP32 using an I2C interface, which reduces the number of required GPIO pins and simplifies wiring. The display provides real-time visual feedback to the user.

6. System Communication and Expansion Capability:

The ESP32 has built-in Wi-Fi and Bluetooth capabilities, allowing future enhancements such as mobile app integration, cloud data storage, or remote monitoring. This makes the system scalable and suitable for IoT-based applications.

7. Calibration of Flex Sensors:

Each flex sensor is calibrated individually to ensure accurate gesture detection. Calibration involves determining the minimum and maximum resistance values for each finger position and defining threshold limits for gesture recognition. Proper calibration improves consistency and accuracy.

8. Data Processing and Output:

The ESP32 continuously reads sensor data, converts analog signals into digital values, and applies logical conditions to identify gestures. Once detected, the system simultaneously displays the corresponding text on the LCD and plays the related audio output.

9. System Testing and Validation:

The complete system is tested using multiple predefined gestures to evaluate accuracy, response time, and synchronization between text and voice output. The results demonstrate reliable real-time performance for assisting communication among speech- and hearing-impaired individuals.

3.6 Flowchart

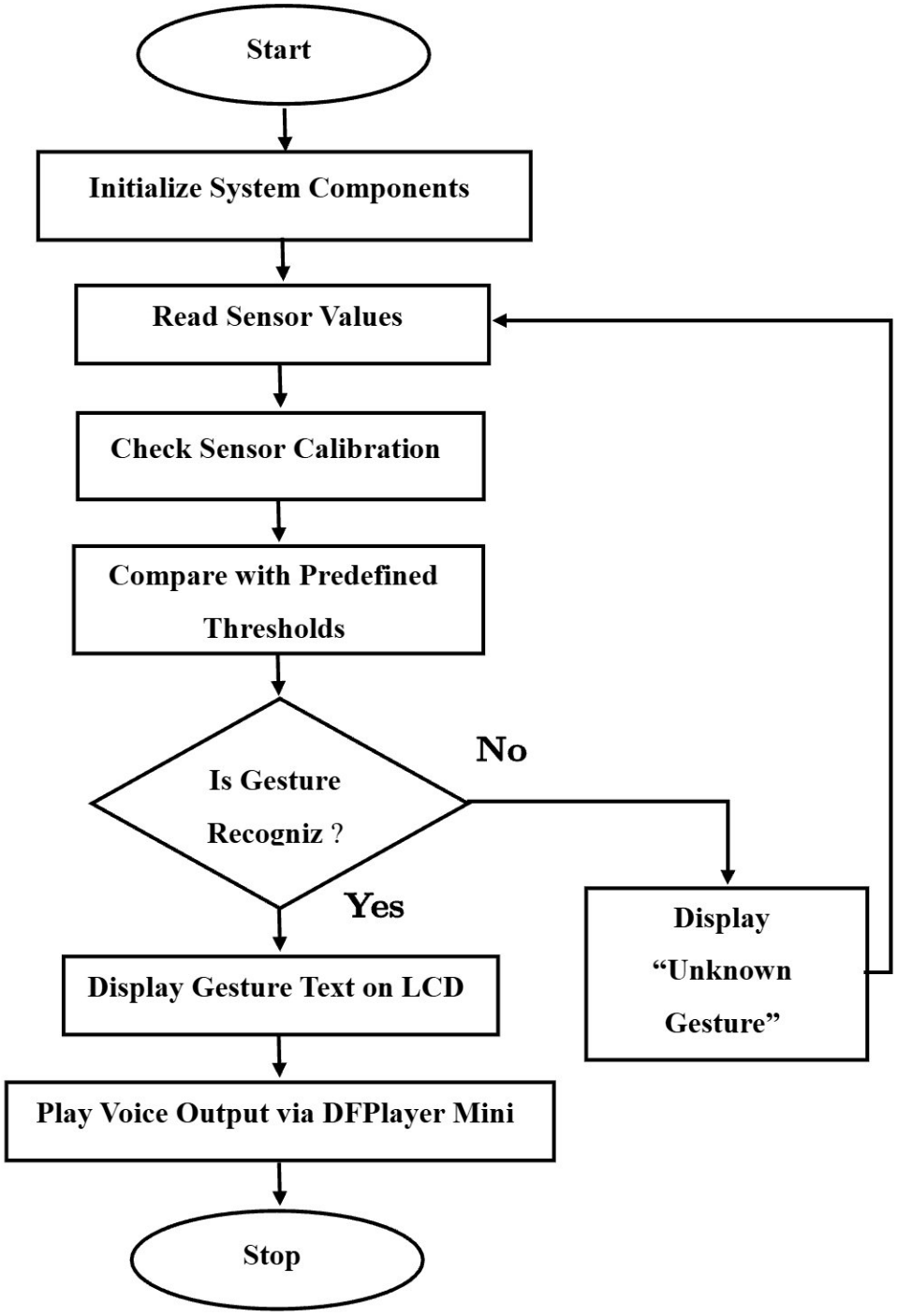

Fig 3.2 Flow Chart

3.7 Detailed Explanation of Flow Chart:

1. Start

The system begins operation when power is supplied to the circuit. The ESP32 initializes all connected components such as the LCD display, flex sensors, and audio module.

2. Initialize LCD and Components

At startup, the ESP32 initializes all peripherals:

- The 16×2 LCD is initialized using the I2C interface for displaying gesture information.
- The DF Player Mini is configured using UART communication for audio playback.
- The ADC channels of the ESP32 are initialized to read analog signals from the\flex sensors.
- The MPU6050 is initialized via I2C for motion detection.

3. Read Sensor Values

The flex sensors detect finger bending and produce analog voltage outputs proportional to the bend. These signals are read by the ESP32 ADC pins and converted into digital values.

If used, the MPU6050 simultaneously provides motion and orientation data.

4. Compare with Threshold Values

The obtained digital values from sensors are compared with predefined threshold limits stored in memory. Each valid combination corresponds to a specific predefined gesture pattern.

5. Gesture Recognized?

This decision block checks whether the current sensor values match any stored gesture:

- If Yes: The gesture is identified successfully, and the system proceeds to output.
- If No: The system continues reading sensor data in a loop.

6. Display Gesture on LCD

Once a gesture is recognized, the corresponding text (e.g., "HELLO", "YES", "NO") is displayed on the 16×2 LCD, providing real-time visual feedback.

7. Play Voice Output

The ESP32 sends a command to the DF Player Mini, which plays the corresponding audio file stored on the SD card through the speaker.

8. Loop for Next Gesture

After completing the output, the system returns to reading sensor values. This loop ensures continuous real-time operation without any manual intervention.

3.8 Block Diagram

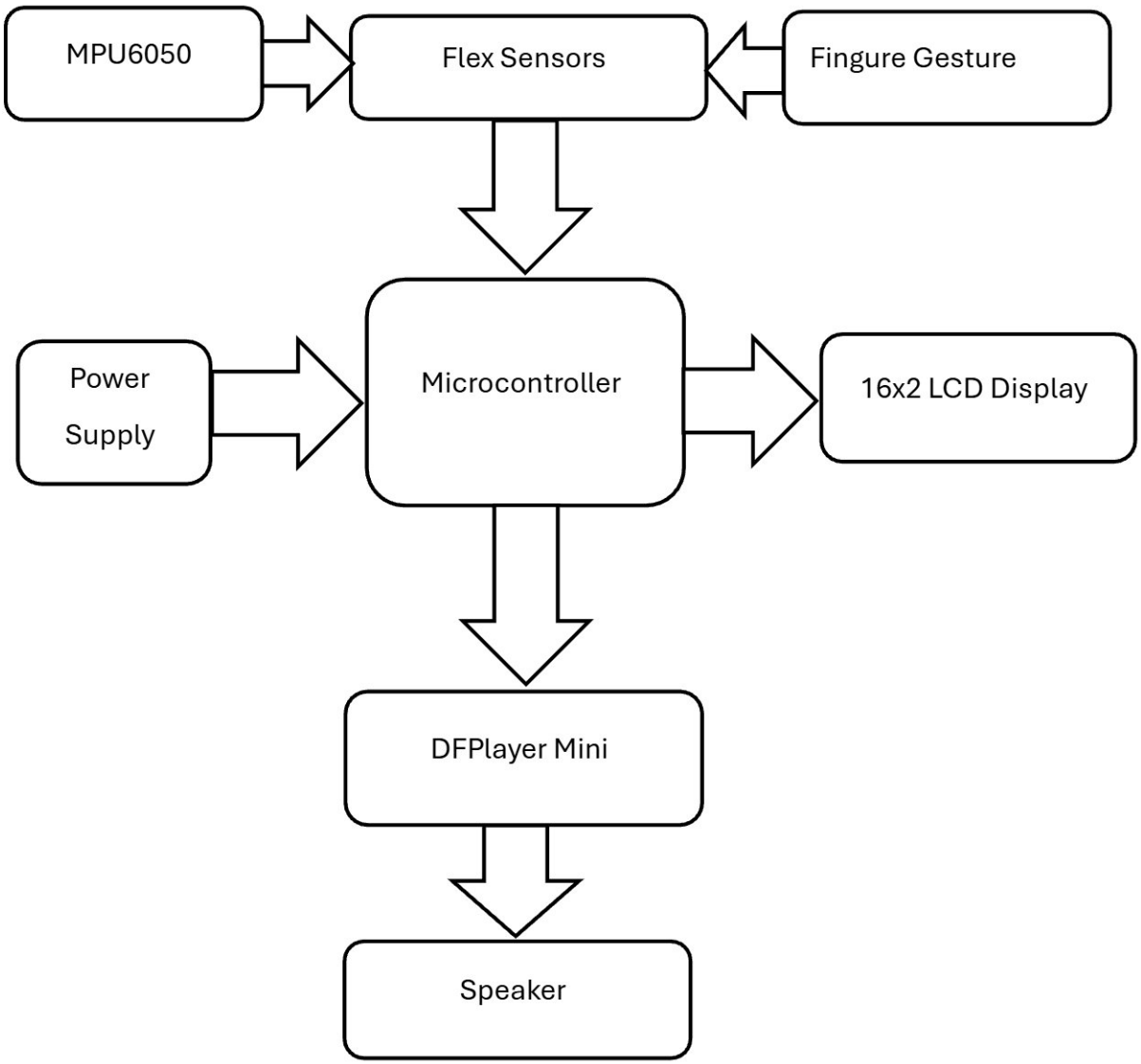

Fig 3.3 Block Diagram

3.9 Detailed Explanation of Block Diagram:

1. Flex Sensors

Flex sensors are mounted on each finger of a glove. These sensors act as variable resistors whose resistance changes when bent. When the user performs a hand gesture, the bending of the fingers changes the resistance of the flex sensors. This variation is converted into a corresponding analog voltage signal using a voltage divider circuit. The analog signals are then sent to the microcontroller for further processing.

2. MPU6050

The MPU6050 sensor is used to detect hand motion and orientation. It consists of a 3-axis accelerometer and a 3-axis gyroscope. The accelerometer measures linear motion, while the gyroscope measures rotational movement of the hand. This helps in identifying dynamic gestures such as hand movement and tilt, thereby improving the accuracy of gesture recognition. The sensor communicates with the microcontroller using the I2C protocol.

3. Microcontroller (ESP32)

The ESP32 acts as the central control unit of the system. It receives analog signals from the flex sensors through its ADC pins and motion data from the MPU6050 via I2C communication. The ESP32 processes these inputs, compares them with predefined threshold values, and identifies the corresponding gesture.

4. 16×2 LCD Display

The 16×2 LCD display is used to show the text representation of the recognized gesture. For example, when a gesture corresponding to "HELLO" is detected, the LCD displays the word "HELLO." This provides immediate visual feedback to the user and helps in verifying correct gesture recognition.

5. DF Player Mini

The DF Player Mini is an audio playback module that stores pre-recorded voice messages in MP3 format on a micro-SD card. When the ESP32 sends a command corresponding to a recognized gesture, the DF Player Mini plays the respective audio file. It acts as the interface between the microcontroller and the speaker for voice output.

6. Speaker

The speaker is used to convert the audio signals from the DFPlayer Mini into audible sound. It produces voice output corresponding to the recognized gesture, enabling effective communication for speech-impaired individuals.

7. Power Supply

The power supply provides the required voltage to all components in the system. The ESP32 is powered using a 5V USB supply, while sensors and modules operate at appropriate voltage levels. A stable power supply ensures proper functioning and reliability of the system.

3.10 Circuit Diagram:

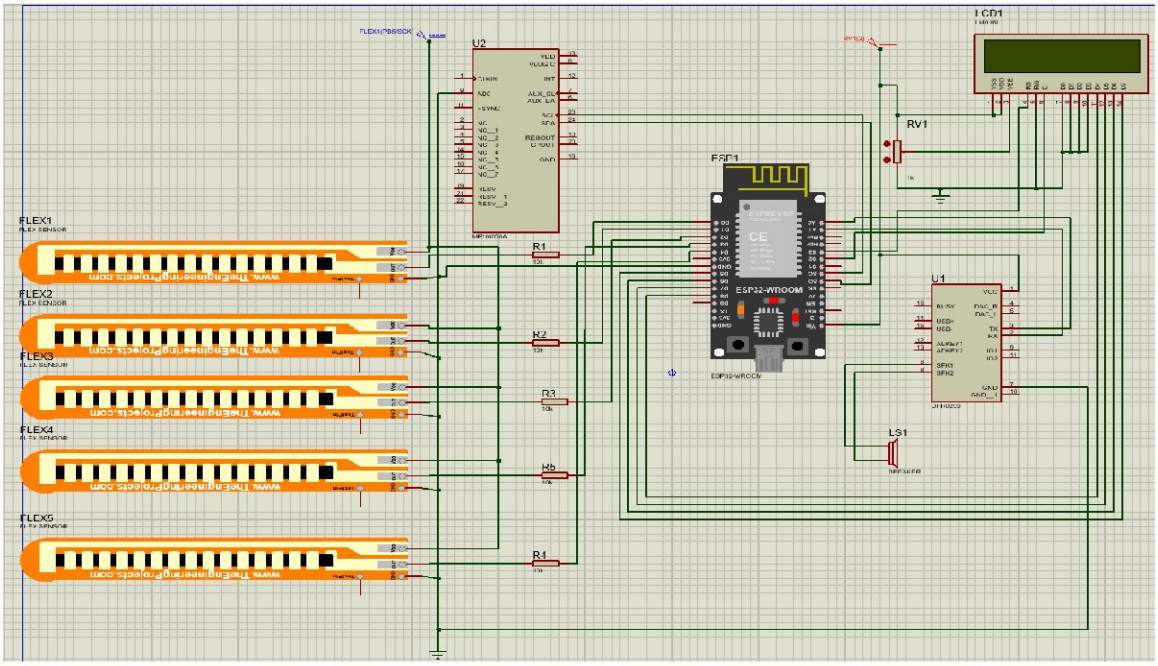

Fig 3.4 Circuit Diagram

3.11 Working

The Sign Language to Voice Conversion system converts hand gestures into both voice and text output using an ESP32. The system uses flex sensors mounted on a glove to detect finger bending. Based on sensor values, predefined gestures are identified and converted into speech and text output.

Detailed Working Process: -

1. Gesture Detection and Signal Generation

- The glove is equipped with five flex sensors, one for each finger.

- When a finger bends, the resistance of the corresponding flex sensor changes.
- This resistance variation produces a corresponding analog voltage signal.
- These analog signals represent different hand gestures and are sent to the ESP32 for processing.

2. Microcontroller Initialization

- The ESP32 is programmed to perform gesture recognition and control output devices.
- On startup, it initializes all connected modules:
    - Configures ADC pins to read flex sensor values
    - Initializes the 16×2 LCD (via I2C)
    - Establishes UART communication with the DF Player Mini Initializes MPU6050 for motion detection
    - After initialization, the system enters continuous monitoring mode.

3. Reading Sensor Data

- Analog signals from the flex sensors are read through ESP32 ADC pins.
- These signals are converted into digital values.
- Each value represents the degree of bending of a finger.
- If used, the MPU6050 simultaneously provides motion and orientation data.

4. Gesture Recognition and Processing

- The ESP32 compares sensor values with predefined threshold ranges stored in memory.
- Each range corresponds to a specific gesture (e.g., “HELLO,” “YES,” “NO”).
- When a match is found, the gesture is recognized.

5. Displaying the Gesture on LCD

- Once identified, the corresponding gesture text is displayed on the 16×2 LCD.
- This provides real-time visual feedback to the user.

6. Voice Output Generation

- Simultaneously, the ESP32 sends a command to the DF Player Mini.
- The DF Player Mini plays a pre-recorded audio file stored on the SD card.
- The output is heard through a speaker.
- Example: If the gesture is “HELLO,” the system plays the audio “HELLO.”

3.12 Software

3.12.1 Proteus

Proteus Design Suite is an Electronic Design Automation (EDA) software tool used for schematic designing, circuit simulation, and verification of electronic systems. It provides an integrated environment for designing and analyzing analog, digital, and microcontroller-based circuits before practical hardware implementation. The software helps engineers and researchers to test circuit functionality, analyze signal flow, and identify possible design errors, thereby reducing development time and improving system reliability. In this project, Proteus was used for the design and simulation of the ESP32-based Sign Language to Voice Converter system. The circuit was developed using various components such as ESP32 microcontroller, flex sensors, MPU6050 accelerometer and gyroscope module, 16×2 I2C LCD display, DFPlayer Mini audio module, speaker, and power supply components. The simulation environment helped in verifying the connections and communication between different modules before implementing the actual hardware prototype.

Proteus enabled the analysis of signal flow between the input sensors, processing unit, and output devices. The flex sensors were analyzed as variable resistance inputs, where changes due to finger bending were converted into voltage variations using voltage divider circuits. These signals were processed by the ESP32 to identify different hand gestures based on predefined threshold values. The MPU6050 sensor interface was also verified to understand the communication between the motion sensor and ESP32 through the I^2C protocol. The simulation

helped in checking sensor integration and ensuring proper data transfer for detecting hand orientation and movement during gesture recognition. Furthermore, Proteus was used to verify the output stages of the system, including the LCD display and voice generation module. The LCD interface was tested to confirm proper display of recognized gesture words, while the DFPlayer Mini module and speaker connection were analyzed for generating the corresponding voice output.

The software provided useful simulation tools such as virtual voltmeters, oscilloscopes, and signal monitoring instruments. These tools helped observe voltage variations from flex sensors, analyze communication signals, and verify the response of the ESP32 controller during different gesture conditions.

Through simulation, component connections and circuit parameters could be modified and optimized before hardware implementation. This reduced troubleshooting time, minimized design errors, and improved the overall reliability of the developed system. The simulation results provided a better understanding of system operation and supported the successful implementation of the practical prototype.

Overall, Proteus played an important role in the development of the Sign Language to Voice Converter system by providing a platform for circuit design, module verification, signal analysis, and performance evaluation. Its simulation capabilities helped ensure proper functioning of the proposed design before final hardware implementation.

3.12.2 Arduino IDE

Arduino Integrated Development Environment (Arduino IDE) is an open-source software platform used for writing, compiling, and uploading programs to microcontroller-based systems. It provides a simple and user-friendly programming environment that allows developers, students, and researchers to create embedded applications efficiently. The software supports various development boards, including Arduino boards and ESP32 microcontrollers, making it suitable for sensor-based, IoT, and automation applications. The availability of built-in libraries and example programs simplifies hardware interfacing and reduces software development time.

In this project, Arduino IDE is primarily used for programming and configuring the ESP32 microcontroller, which acts as the main processing unit of the Sign Language to Voice Converter system. The complete program is developed using Embedded C/C++ programming language and uploaded to the ESP32 through the Arduino IDE environment. The software enables integration of different hardware components such as five flex sensors, MPU6050 accelerometer and gyroscope module, 16×2 I2C LCD display, DFPlayer Mini audio module, and speaker in an efficient manner.

The Arduino IDE provides support for various external libraries required for interfacing different modules. Libraries such as Wire.h, LiquidCrystal_I2C.h, MPU6050 library, and DFRobotDFPlayerMini.h are used for communication with the I2C-based sensors, LCD display, motion sensor, and audio playback module. These libraries provide predefined functions for sensor data acquisition, display control, gesture processing, and voice output generation, which simplifies the overall programming process.

The software is used to implement the gesture recognition algorithm of the system. The analog values obtained from the flex sensors are continuously read through the ESP32 ADC pins, while the MPU6050 provides hand orientation and motion data through I^2C communication. The collected sensor values are compared with predefined threshold values stored in the program to identify specific hand gestures. Once a gesture is recognized, the ESP32 generates the corresponding output command. Based on the detected gesture, the Arduino program controls the LCD display to show the recognized word and sends a command to the DFPlayer Mini module to play the corresponding pre-recorded voice message stored on the micro SD card. This allows conversion of sign language gestures into both text and voice output in real time.

One of the important features used during the development process is the Serial Monitor provided by Arduino IDE. It allows real-time observation of flex sensor readings, MPU6050 values, and system responses during testing and calibration. This feature helps in debugging errors, adjusting threshold values, and improving gesture recognition accuracy.

The Arduino IDE also allows repeated modification and testing of the program, enabling optimization of system performance and reliability. The software environment helps in managing libraries, compiling code, uploading firmware, and verifying communication between ESP32 and connected hardware modules.

The use of Arduino IDE plays an important role in converting the proposed sign language recognition concept into a functional embedded system. By providing an efficient platform for programming, debugging, sensor interfacing, and output control, Arduino IDE contributes significantly to the successful development of the ESP32-based Sign Language to Voice Converter system.

Chapter 4

Hardware Design

4.1 Microcontroller: ESP32

The ESP32 is a powerful 32-bit microcontroller developed by Espressif Systems. It features a dual-core processor operating at up to 240 MHz, along with sufficient RAM and flash memory for handling real-time embedded applications. The ESP32 includes built-in Analog-to-Digital Converters (ADC), digital I/O pins, timers, and multiple communication interfaces such as UART, I2C, and SPI. It also has integrated Wi-Fi and Bluetooth capabilities, making it highly suitable for IoT-based and wireless applications.

Fig 4.1 Microcontroller (ESP32)

Reason of selection:

- **Built-in ADC:** The 12-bit Analog-to-Digital Converter allows accurate reading of analog signal.
- **High Performance:** Operates at up to 240 MHz, ensuring fast processing and real-time response.
- **Multiple GPIO Pins:** Provides sufficient pins to interface with sensors, LCD, and communication modules.
- **Multiple Communication Interfaces:** Supports UART, I2C, and SPI, enabling easy interfacing with modules like LCD, MPU6050, and DF Player.
- **Wi-Fi & Bluetooth Support:** Built-in wireless capability allows future upgrades such as mobile app integration or IoT-based monitoring.
- **Cost-Effective and Versatile:** Widely used, affordable, and suitable for both beginner and advanced embedded applications.

Pin Diagram of ESP32:

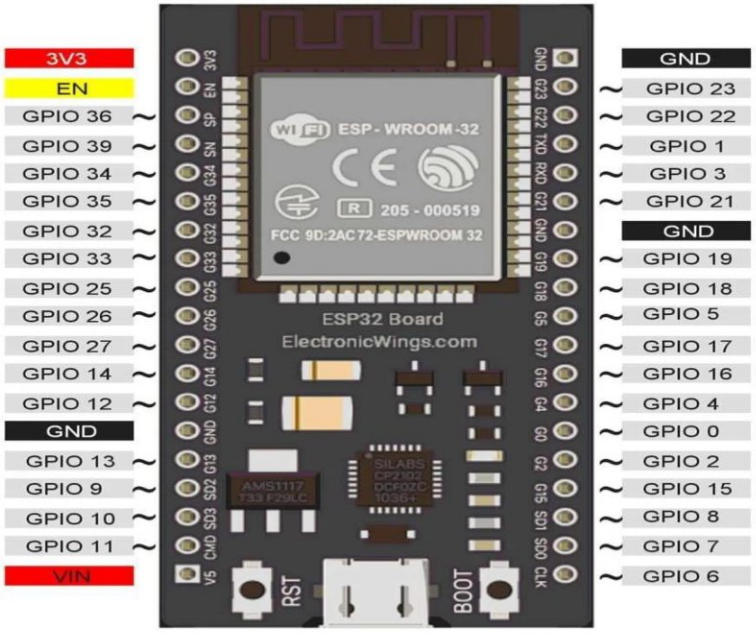

Fig 4.2 ESP32 Pin Diagram

4.2 Flex Sensor

A flex sensor is a variable resistor that changes its resistance according to the amount of bending applied to it. It is a thin, flexible sensor commonly used in wearable electronics and gesture recognition applications. In this project, five flex sensors are attached to the fingers of a glove to detect individual finger movements during sign language gestures. When the user bends a finger, the flex sensor also bends, causing a change in its internal resistance. The resistance value increases as the bending angle increases. This change in resistance is converted into a voltage variation using a voltage divider circuit. The analog voltage signal is then provided to the ADC pins of the ESP32 for processing and gesture identification. The ESP32 continuously reads the analog values from all five flex sensors and compares them with predefined threshold values stored in the program. Based on the combination of sensor values, the system identifies the performed hand gesture and generates the corresponding text and voice output.

Construction and Working Principle of Flex Sensor:

A flex sensor is a flexible variable resistor used to detect bending movement. It consists of a thin plastic substrate coated with a carbon-based resistive material. The resistance of the sensor changes according to the bending angle.

- Straight Condition: When the sensor is straight, the conductive layer remains compact, allowing easy current flow. The resistance is low, approximately 10 kΩ.
- Bent Condition: When the sensor is bent, the conductive layer stretches, increasing the resistance due to reduced conductivity. The resistance can increase up to 30–50kΩ depending on the bending angle.

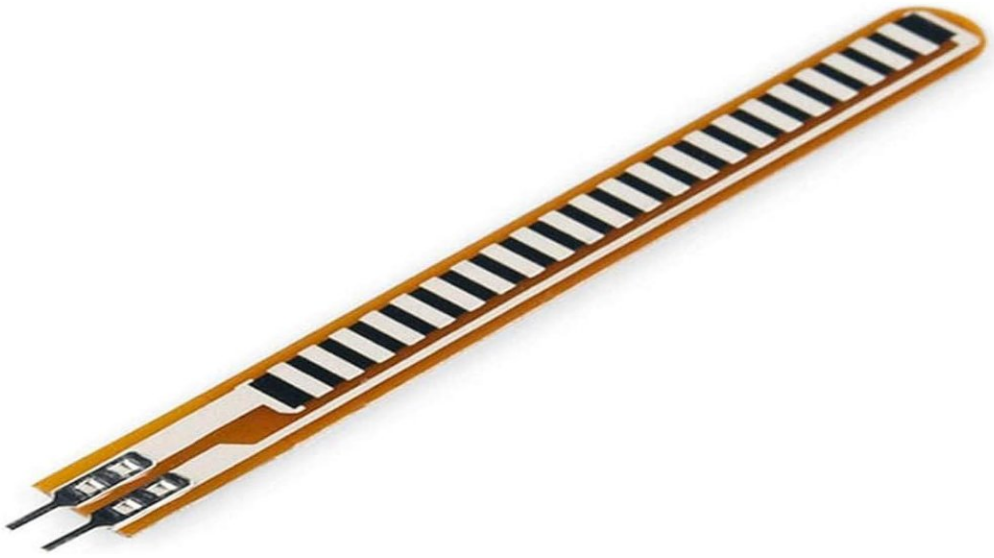

Fig 4.3 Flex sensor

Flex Sensor Voltage Calculation (Voltage Divider):

Flex sensor is connected with a 10 kΩ resistor.

Formula:

$$V_{out} = V_{cc} \times \frac{R_{fixed}}{R_{flex} + R_{fixed}}$$

- $V_{cc} = 3.3V$
- $R_{fixed} = 10k\Omega$
- $R_{flex} = 10k\Omega$(straight) to $30k\Omega$(bent)

Case 1: Finger Straight (10kΩ)

$$V_{out} = 3.3 \times \frac{10}{10 + 10} = 3.3 \times 0.5 = 1.65V$$

Case 2: Finger Bent (30kΩ)

$$V_{out} = 3.3 \times \frac{10}{30 + 10} = 3.3 \times 0.25 = 0.825V$$

Specifications:

Sr. No	Parameter	Typical Value / Range
1.	Length	2.2 inch or 4.5 inch
2.	Flat Resistance	10 kΩ
3.	Resistance when Bent (90°)	30–50 kΩ
4.	Bending Angle Range	0° to 90°
5.	Operating Voltage	0–5V DC
6.	Power Consumption	Very low
7.	Output Signal	Analog voltage proportional to bend
8.	Life Cycle	>1 million bend operations

Table 4.1 Specification

Current Calculation of Flex Sensor:

$$I = \frac{V}{R}$$

Given:

- Supply Voltage = 3.3V
- Fixed Resistor = 10kΩ
- Flex Sensor Resistance = 10kΩ

$$R_{total} = 10k\Omega + 10k\Omega = 20k\Omega$$

$$I = \frac{3.3}{20000} = 0.000165A$$

4.3 16x2 LCD Display:

The 16x2 Liquid Crystal Display (LCD) is a widely used module for displaying alphanumeric characters. It consists of two rows, each capable of displaying 16 characters, making it ideal for showing real-time text output such as recognized gestures.

Specifications:

- Display Type: Alphanumeric LCD
- Display Format: 16 characters × 2 lines
- Operating Voltage: 4.7V to 5.3V (typically 5V)
- Interface: Parallel or I2C (two-wire communication)
- Backlight: LED for better visibility
- Characters: 5×8 dot matrix for each character

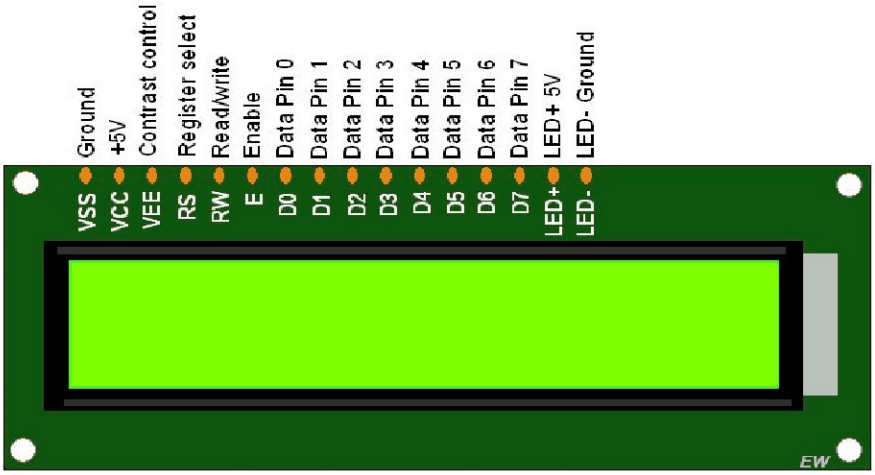

Fig 4.4 LCD Display

4.4 DF Player Mini

The DFPlayer Mini is a compact and low-cost MP3 audio playback module used for generating voice output in embedded systems. It can directly play audio files stored on a micro SD card and is controlled by a microcontroller through serial communication. In this project, the DFPlayer Mini is used to convert the recognized sign language gesture into a voice message by playing the corresponding pre-recorded audio file through a speaker. The ESP32 sends commands to the DFPlayer Mini according to the detected hand gesture. The module searches for the corresponding audio file stored in the micro SD card and plays it through the connected speaker, providing real-time voice output for the user.

Working Principle:

- The voice messages corresponding to different sign language gestures are stored in a micro SD card in MP3 format.
- The ESP32 sends a serial command to the DFPlayer Mini through the UART communication interface.
- The DFPlayer Mini processes the command and reads the selected audio file from the SD card.
- The built-in audio decoder converts the digital audio file into an analog audio signal.

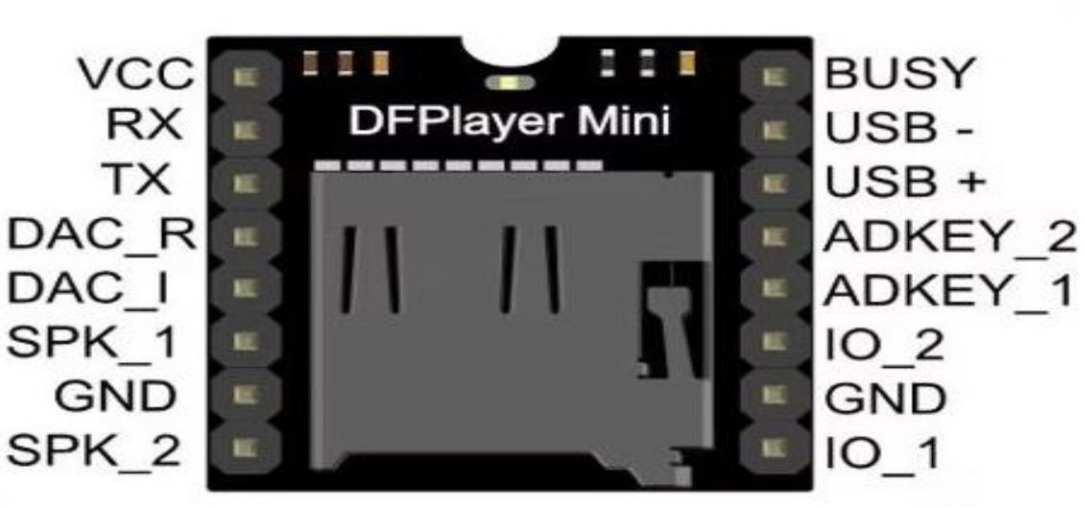

Fig 4.5 DF Player Mini

4.5 MPU6050

The MPU6050 is a motion sensing module that combines a 3-axis accelerometer and a 3-axis gyroscope in a single chip. It is used to detect the movement, tilt, and orientation of the hand during gesture formation. The accelerometer measures linear acceleration along the X, Y, and Z axes, while the gyroscope measures angular rotation around these axes. This sensor helps in recognizing dynamic gestures by providing motion data, which improves the overall accuracy of the system. The MPU6050 communicates with the microcontroller using the I2C protocol.

Working Principle:

- The accelerometer measures the acceleration of the hand in three directions (X, Y, and Z axes).
- The gyroscope measures the angular rotation and orientation of the hand.

- The MPU6050 converts physical motion into digital signals using its internal 16-bit ADC.
- The ESP32 receives and processes this motion data along with flex sensor values.
- The combined sensor data helps identify different hand gestures accurately.

Specifications of MPU6050:

- Sensor Type: 6-Axis Motion Sensor (3-axis Accelerometer + 3-axis Gyroscope)
- Operating Voltage: 3.3V to 5V
- Interface: I²C (two-wire communication)
- Accelerometer Range: ±2g, ±4g, ±8g, ±16g
- Gyroscope Range: ±250°/s, ±500°/s, ±1000°/s, ±2000°/s
- Resolution: 16-bit ADC
- Power Consumption: Approximately 3.5 mA
- Application: Motion detection, gesture recognition, and orientation sensing.

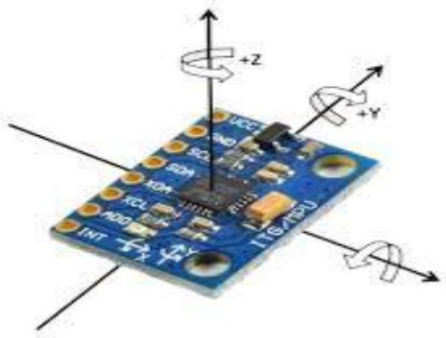

Fig 4.6 MPU6050

4.6 Speaker

A speaker is an output audio device used to convert electrical audio signals into sound waves that are audible to humans. In this project, the speaker is used to produce the voice output corresponding to the recognized sign language gesture. When a gesture is detected and processed by the microcontroller, a command is sent to the DF Player Mini module, which plays the appropriate pre-recorded audio file through the speaker.

Fig 4.7 MPU6050

4.7 Li-ion Battery

The Li-ion battery is a rechargeable power source used to provide portable power to the sign language conversion system. It allows the glove-based system to operate independently without requiring a continuous external power supply. The battery supplies power to the ESP32, sensors, LCD display, and audio module through suitable power regulation circuits.

Specifications:

- Nominal Voltage: 3.3V DC
- Capacity: 2200mAh • Output Voltage: 3.3V

4.8 Bill of Materials (BOM)

Sr. No.	Component Name	Specification	Quantity	Approx Cost (₹)
1	ESP32 Development Board	ESP32 Wi-Fi + Bluetooth, 3.3V Logic	1	400
2	Flex Sensors	2.2 inch bend sensor, 10kΩ–30kΩ	5	2500
3	MPU6050 Module	6-axis Accelerometer + Gyroscope	1	150
4	16×2 LCD Display	I2C Interface	1	250
5	DFPlayer Mini Module	MP3 Audio Playback Module	1	200
6	Speaker	8Ω, 3W	1	150
7	Micro SD Card	8GB FAT32	1	300
8	Resistors	10kΩ, ¼ Watt	10	50
9	Battery	3.7V Li-ion Rechargeable	1	300
10	Breadboard & Jumper Wires	Connecting Accessories	1 Set	300
11	Glove	Fabric Glove for Sensor Mounting	1	100
12	USB Cable	Micro USB	1	150

Table 4.2 Bill Of Materials

Chapter 5

Conclusion and Future Scope

5.1 Conclusion:

The "Sign Language to Voice Converter using ESP32" is a low-cost and efficient system that converts predefined hand gestures into text and voice output. Using flex sensors and the DF Player Mini, the system provides real-time communication support for speech- and hearing-impaired individuals. Overall, it is a portable, reliable solution that helps bridge the communication gap and can be further enhanced in the future.

Once completed, the system is expected to:

- Accurately detect predefined hand gestures.
- Display recognized gestures on a 16x2 LCD screen.
- Generate corresponding voice output using the DF Player Mini.
- Provide a user-friendly, low-cost, and portable communication aid.

5.2 Future Scope:

While the current prototype performs well for a limited set of gestures, there are several opportunities to enhance its performance and usability in the future:

1. Integration of Machine Learning / AI: Future versions can include machine learning algorithms to recognize a larger set of gestures and even dynamic (moving) signs for complete sign language translation.

2. Wireless Connectivity Enhancement: The system can be upgraded with Bluetooth or Wi-Fi modules for real-time data transmission to smartphones, IoT platforms, or healthcare servers.
3. Mobile Application Interface: A dedicated Android/iOS app can be developed to display recognized text, store gesture history, and offer text-to-speech conversion for better accessibility.
4. Text-to-Speech (TTS) System Integration: Instead of using pre-recorded MP3 files, real-time text-to-speech engines can be implemented for more flexible and natural communication.
5. Improved Sensor Accuracy: Advanced bend sensors or IMU sensors (Inertial Measurement Units) can be used to detect complex hand movements with higher accuracy and speed.
6. Multilingual Voice Support: The audio system can be expanded to support multiple languages, allowing users to communicate in their preferred regional or national language.

Chapter 6
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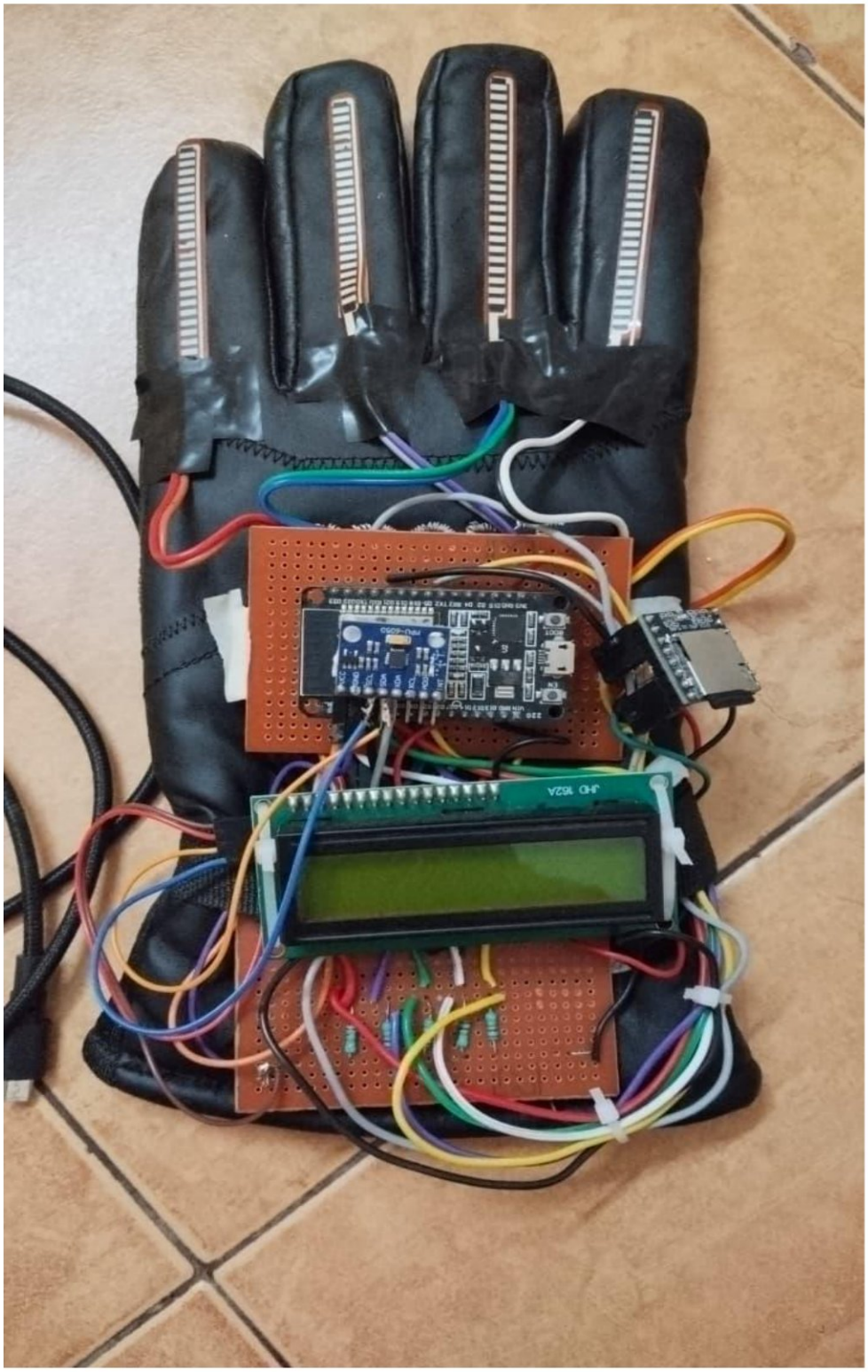

Developed Prototype of Sign Language to Voice Converter System

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2. Published Paper

SIGN LANGUAGE TO VOICE CONVERTER

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Abstract

Digital Vocalizer is a project aimed at bridging the communication gap between mute and hearing-impaired individuals and the general public. Deaf and mute individuals commonly rely on sign language or hand gestures for communication, which can be difficult for others to interpret. To overcome this challenge, the proposed system uses an **ESP32 microcontroller** as the main control unit to interface various sensors and output devices. The system is based on the concept of gesture recognition using sensor data. **Flex sensors** are mounted on a glove to detect finger bending, while the **MPU6050 sensor** is used to measure hand orientation and motion. These sensor outputs are converted into electrical signals and processed by the ESP32 using its ADC with the help of a **10kΩ resistor-based voltage divider circuit**. The controller interprets the gesture patterns and converts them into corresponding text displayed on a **16×2 LCD** and speech output using a **DFPlayer Mini module** connected to a speaker and SD card. This system enables real-time communication, improving accessibility and interaction for speech- and hearing-impaired individuals.

Keywords: Flex Sensor, MPU6050, ESP32, Voltage Divider, DF Mini Player Mini, LCD, Voice Output communication gap.

I. INTRODUCTION

Digital Vocalizer system is designed as a smart solution that translates hand gestures into both text and speech **flex sensors** to Communication is a fundamental aspect of in real time. The system uses human interaction, **MPU6050** enabling individuals to detect finger movements and an express thoughts, ideas, and emotions effectively. However, for speech- and hearing-impaired sensor to capture hand orientation and motion. **ESP32** These inputs are processed by the individuals, communication with the general public remains a major challenge. They primarily **microcontroller**, which acts as the core processing unit. The detected gestures are relied on sign language and hand gestures, which interpreted using predefined **16×2** patterns, and they are not widely understood by everyone. This corresponding output is displayed on an often leads to difficulties in daily interactions, social isolation, and reduced independence. With **LCD** while also being converted into voice using a **DFPlayer Mini module** connected to a speaker. Advancements in embedded systems and sensor

assistive devices that can bridge this knowledge of sign language. The proposed system focuses on portability, cost-effectiveness, and ease of use, making it a practical and efficient assistive technology. By combining sensor-based gesture recognition with real-time voice output, the Digital Vocalizer enhances communication, promotes inclusivity, and improves the overall quality of life for individuals with disabilities.

II.METHODOLOGY

The methodology of the Digital Vocalizer system is based on sensor-based gesture recognition and real-time signal processing using an embedded system. The complete working process is divided into several stages, including data acquisition, signal processing, gesture recognition, and output generation. Initially, **flex sensors** are mounted on the fingers of a glove to detect bending movements. As the fingers bend, the resistance of the flex sensors changes, which is converted into voltage variations using a **10kΩ resistor-based voltage divider circuit**. Along with this, the **MPU6050 sensor** is used to capture hand orientation and motion such as tilt and acceleration. These sensors collectively provide accurate gesture data. The generated analogy signals from the flex sensors are fed into the **ESP32 microcontroller**, where they are converted into digital values using the built-in ADC. The MPU6050 provides digital data through I2C communication. The ESP32 continuously reads and processes these inputs in real time. Next, the system compares the obtained sensor values with predefined threshold patterns stored in the program. Based on this comparison, the system identifies the corresponding hand gesture. Once the gesture is recognized, the microcontroller triggers the appropriate output. The recognized gesture is then displayed as text on a **16×2 LCD display** for visual plays a This enables seamless interaction between deaf or technologies, there growing opportunity to mute individuals and others without requiring develop prerecorded audio file stored in the SD card through a speaker, converting the gesture into speech.

2.1 Motivation:

Communication barriers faced by speech- and hearing-impaired individuals create significant challenges in everyday life, as most people are not familiar with sign language. This often leads to difficulties in expressing basic needs, reduced social interaction, and increased dependency on others. Therefore, there is a strong need for a system that can simplify communication and make it understandable for everyone. The motivation behind the Digital Vocalizer project is to develop a cost-effective, portable, and real-time solution that converts hand gestures into speech and text. By using modern technologies like ESP32, flex sensors, and MPU6050, the system aims to provide an easy-to-use assistive device. This helps improve independence, confidence, and overall quality of life for speech- and hearing-impaired individuals while promoting inclusivity in society.

2.1 Objectives:

1. To design and develop a system that converts sign language gestures into audible speech in real time.
2. To use flex sensors and MPU6050 for accurate detection of finger movements and hand orientation.
3. To implement the ESP32 microcontroller for efficient data processing and gesture recognition.
4. To display the recognized gestures as text on a

5. To create a portable, cost-effective, and user-friendly assistive device that improves communication for speech- and hearing-impaired individuals.

2.3Literature Review:

1.Smart Glove for Sign Language Recognition Using Flex Sensors and Arduino UNO Authors: A.Sharma, R.Singh,and P.Deshmukh Journal: International Journal of Innovative Research in Computer and Communication Engineering (IJIRCCE), 2019

Abstract: This paper presents the design of a smart glove equipped with flex sensors to recognize hand gestures corresponding to sign language alphabets. The system uses an Arduino UNO microcontroller to process the sensor signals and displays the recognized gesture on an LCD screen. It provides a simple and cost-effective solution for basic sign language interpretation.

2.IoT-Based Sign Language Translation System Using ESP32 Authors: S.Patel, M.Reddy, and A.Ghosh Journal: International Journal of Advanced Research in Electronics and Communication Engineering (IJARECE), 2021

Abstract: The paper introduces an IoT-enabled system for converting sign language gestures into text and speech using an ESP32 microcontroller. Flex sensors capture the gesture data, and the ESP32 transmits the recognized text over Wi-Fi to a web dashboard. The 6 recognized gesture is also converted into voice output using an MP3 playback module. **3. Real-Time Gesture to Speech**

DFPlayer Mini module. Commands to the **DFPlayer Mini module**, which

Conversion Using Flex Sensors Authors: R. Joshi, K. Bansal, and D. Mehta Journal: International Journal of Scientific Research in Engineering and Technology (IJSRET), 2020

Abstract: This research focuses on the development of a gesture-based communication system using flex sensors and a microcontroller. The system identifies predefined hand gestures and plays corresponding voice messages through a speaker module. It aims to assist speech-impaired individuals by providing Realtime gesture-to-speech conversion.

4.Design and Implementation of Smart Communication Glove Using Flex Sensors and 7 Wi-Fi Module Authors: N. Kumar, V. Sharma, and T. Pandey **Journal:** Journal of Embedded Systems

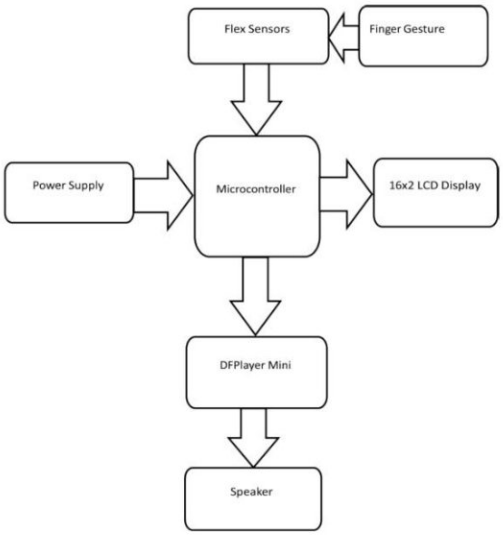
5.IoT-Based Assistive Glove for Sign to Speech Conversion Using ESP32 and DFPlayer Mini

Authors: P. Gupta, H. Thakur, and S. Verma Journal: International Journal of Emerging Technologies and Innovative Research (JETIR), 2023

Abstract: The authors propose a complete IoT-based sign-to-speech system that utilizes an ESP32 microcontroller, flex sensors, and DFPlayer Mini for audio output. The device converts detected gestures into corresponding speech and sends recognized text data to a cloud dashboard using Wi-Fi. An LCD display provides instant visual output to the user.

III. ANALYSIS AND DESIGN APPROACH

3.1 System Architecture



1. ESP32 Microcontroller

The ESP32 is the main controller of the system, responsible for processing sensor data and controlling all operations. It has built-in Wi-Fi and Bluetooth capabilities, high processing speed, and multiple ADC channels. In this project, it reads data from flex sensors and the MPU6050, processes gesture patterns, and controls the LCD and DFPlayer Mini module.

2. Flex Sensors

Flex sensors are used to detect the bending of fingers. Their resistance changes when they are bent, which helps in identifying different hand gestures. These sensors are mounted on a glove and play a crucial role in capturing finger movement data for gesture recognition.

3. MPU6050 Sensor

The MPU6050 is a motion tracking device that combines a 3-axis accelerometer and a 3-axis gyroscope. It is used to detect hand orientation, tilt, and motion. This improves the accuracy of gesture detection by providing additional movement data.

4.16×2 LCD Display

The 16×2 LCD is used to display the recognized gesture in text format. It provides visual feedback to the user, making the system more interactive and user-friendly.

5.DFPlayer Mini (BF Mini Player) The DFPlayer Mini is a compact audio module used for voice output. It reads audio files stored on an SD card and plays them through a speaker. It is controlled by the ESP32 to produce speech corresponding to detected gestures.

6.Speaker

The speaker is connected to the DFPlayer Mini to output the audio signals. It converts electrical signals into sound, allowing the system to communicate gestures as audible speech.

7.SD Card

The SD card is used to store pre-recorded audio files. Each file corresponds to a specific gesture, which is played when the gesture is recognized.

3.2Advantages:

1.Improves Communication: Converts sign language into speech and text, making it easier for others to understand.

1. **Cost-Effective:**
   Uses affordable components like ESP32 and sensors, making it economical.
2. **Dual Output (Text + Voice):**
   Displays text on LCD and produces audio via speaker for better understanding.
3. **Improved Accuracy:**
   Combination of flex sensors and MPU6050 enhances gesture recognition.
4. **Promotes Independence:**
   Reduces dependency on interpreters or others for communication.

3.3Applications:

1. **Assistive Communication Device:** Helps speech- and hearing-impaired individuals communicate effectively with others in daily life.
2. **Hospitals and Healthcare:** Enables patients with disabilities to convey their needs easily to doctors and nurses.
3. **Educational Institutions:** Assists students with hearing or speech impairments in interacting with teachers and classmates.
4. **Public Places:** Useful in places like banks, railway stations, and offices where quick communication is required.
5. **Customer Service:** Can be used in service sectors to help differently abled employees interact with customers.
6. **Smart Wearable Technology:** Acts as a wearable device (glove-based system) for real-time gesture-to-speech conversion.
7. **Research and Development:** Useful for further research in embedded systems.

3.4Conclusion:

The Digital Vocalizer system successfully addresses the communication gap between speech- and hearing-impaired individuals and the general public by converting hand gestures into both text and audible speech. By integrating flex sensors, MPU6050, and the ESP32 microcontroller, the system is able to accurately detect and interpret gestures in real time. The use of a DFPlayer Mini module and speaker ensures clear voice output, while the 16×2 LCD provides visual feedback, making the system more effective and user-friendly. Overall, the project demonstrates a cost-effective, portable, and efficient solution for assistive communication. It enhances independence, improves social interaction, and promotes inclusivity for differently-abled individuals. With further improvements such as advanced gesture recognition.

IV. ACKNOWLEDGEMENT

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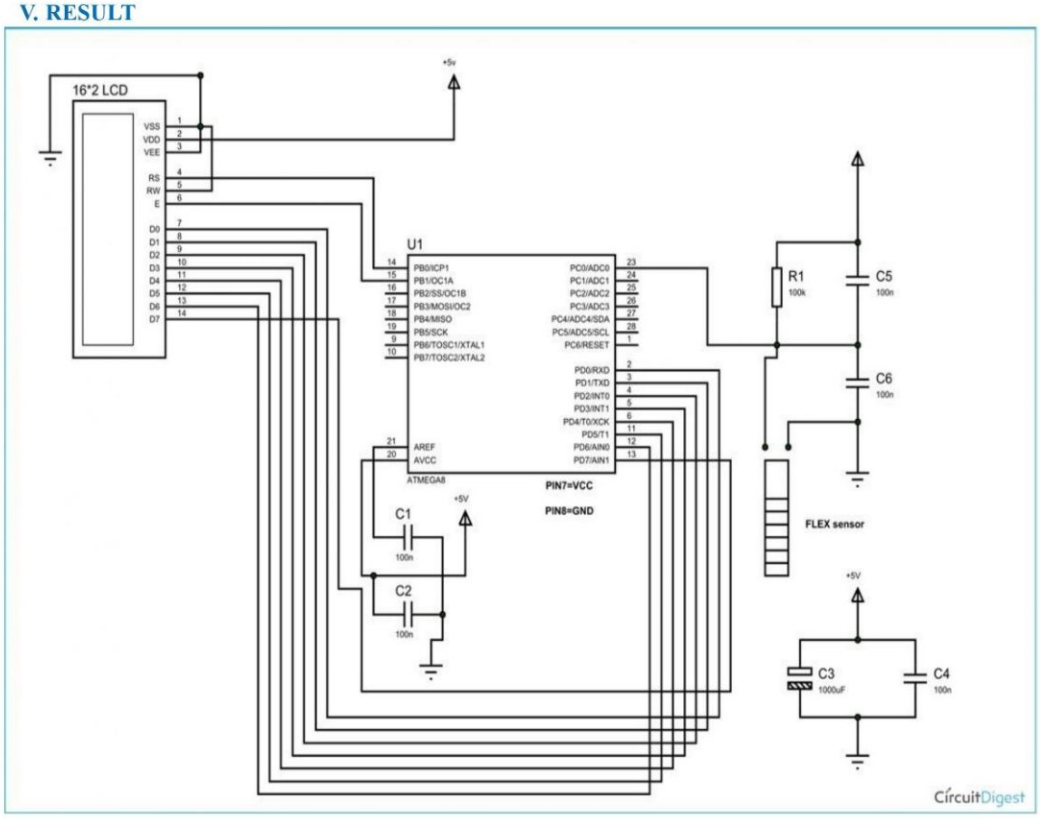
V. RESULT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PB7/TOSC2/XTAL2
AREF
AVCC
ATMEGA8
PC0/ADC0
PC1/ADC1
PC2/ADC2
PC3/ADC3
PC4/ADC4/SDA
PC5/ADC5/SCL
PC6/RESET
PD0/RXD
PD1/TXD
PD2/INT0
PD3/INT1
PD4/T0/XCK
PD5/T1
PD6/AIN0
PD7/AIN1
PIN7=VCC
PIN8=GND
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C6
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FLEX sensor
C3
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C4
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CircuitDigest

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SIGN LANGUAGE TO VOICE CONVERTER
Chapter 1 Introduction
1.1 Necessity:
Communication plays a crucial role in human interaction, but individuals with speech and hearing disabilities often struggle to express their thoughts effectively. Since most people are not familiar with sign language, a significant communication gap exists between speech-impaired individuals and the general population. Therefore, there is a strong need for a system that can automatically convert hand gestures into speech or text. The proposed Sign Language to Voice Converter using the ESP32 aims to fulfil this necessity. By using flex sensors attached to a glove, the system detects finger movements corresponding to different sign gestures. Additionally, a MPU6050 is used to detect hand motion and orientation, improving the accuracy of gesture recognition. These signals are processed by the ESP32, which interprets them and generates the corresponding text on an LCD display and voice output using the DF Player Mini. This enables smooth and effective communication between speech-impaired individuals and others.
1.2 Need of Project:
People with speech and hearing impairments face major challenges when communicating with those unfamiliar with sign language. To assist them, there is a need for a low-cost, portable, and reliable device that can recognize hand gestures and translate them into meaningful speech or text in real time. The Sign Language to Voice Converter using the ESP32 fulfils this requirement by converting the analog data from flex sensors into digital signals using the built-in ADC (Analog-to-Digital Converter) of the ESP32. In addition, a MPU6050 is used to capture hand movement and orientation, which improves gesture recognition accuracy. These processed signals are then used to generate audible voice output through the DF Player Mini and display the corresponding text on an LCD screen, thereby bridging the communication gap and promoting inclusivity in society.
1
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1.3 Motivation:
SIGN LANGUAGE TO VOICE CONVERTER
Communication is an essential aspect of human life, but individuals with speech and hearing impairments often face significant challenges in expressing their ideas and emotions. They primarily rely on sign language, which is not widely understood by the general population. This communication gap creates barriers in education, healthcare, and daily social interactions. This limitation inspired the development of a Sign Language to

Voice Converter using the ESP32, which can interpret hand gestures and convert them into both speech and text. The project aims to design an affordable, reliable, and portable assistive device using flex sensors along with a MPU6050 for improved gesture detection, and voice playback technology using the DF Player Mini. The main motivation behind this work is to utilize embedded system design and modern microcontroller technology to create an innovative solution that enhances social inclusion and enables differently-abled individuals to communicate independently and confidently.

1.4 Problem Statement:

Communication between deaf and mute individuals and others is difficult due to limited understanding of sign language. Existing solutions are often complex or expensive. There is a need for a simple system that can recognize predefined hand gestures and convert them into voice. This project uses sensors and an ESP32 to provide real-time gesture-to-speech conversion.

1.5 Objectives:

• To design a wearable glove integrated with flex sensors to detect finger movements representing predefined sign language gestures.

• To use an ESP32 for acquiring and processing flex sensor data and identifying specific gesture patterns.

• To integrate a motion sensor such as MPU6050 for detecting hand movement and improving gesture recognition accuracy.

• To convert recognized gestures into speech output using a DF Player Mini and speaker for effective audio communication.

2

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SIGN LANGUAGE TO VOICE CONVERTER

• To display the identified gesture text on a 16×2 LCD using the I2C interface for real-time visual feedback.

• To develop a low-cost, reliable, and portable assistive system that bridges the communication gap between sign language users and non-signers.

1.6 Advantages and Limitations:

1.6.1 Advantages:

• Assistive Communication: Enables speech- and hearing-impaired individuals to communicate effectively through gesture-based voice conversion.

• Low Cost and Accessibility: Built using affordable and easily available components like ESP32, flex sensors, and LCD, making the system economical and easy to implement.

• Compact and Portable Design: Lightweight, glove-based system that can be easily carried and used in daily life.

• Easy Interfacing: The ESP32 supports multiple interfaces (ADC, I2C, UART), allowing simple and efficient integration with sensors, LCD, and DF Player Mini.

• Improved Accuracy: Integration of motion sensing using MPU6050 enhances gesture detection by combining finger movement and hand orientation.

• Upgradeable System: The system can be further enhanced using machine learning, additional sensors, or wireless connectivity for advanced gesture recognition.

• Educational Value: Demonstrates important embedded system concepts such as ADC conversion, sensor interfacing, I2C communication, and serial communication.

3

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1.6.2 Limitations:

SIGN LANGUAGE TO VOICE CONVERTER

• Sensor Calibration Required: Flex sensors must be properly calibrated to ensure accurate gesture detection and to avoid incorrect readings.

• Limited Gesture Recognition: The system recognizes only a predefined set of gestures and does not support complete sign language vocabulary or dynamic gesture learning.
• Display Constraints: The 16×2 LCD can display only a limited number of characters, restricting the amount of information shown at a time.
• Pre-recorded Audio Output: Voice output depends on stored audio files in the DF Player Mini, rather than real-time text-to-speech conversion.
• User Variation: Accuracy may vary depending on hand size, finger flexibility, and differences in how users perform gestures.
• Wired Prototype Design: The use of wired connections and external modules may reduce comfort and flexibility for long-term wearable use.

1.7 Applications:

1. Assistive Device for Disabled: Helps speech- and hearing-impaired individuals communicate effectively through gesture-to-voice translation.
2. Educational Institutions: Can be used in special schools to teach and demonstrate gesture-based communication systems.
3. Healthcare Sector: Useful for patients with speech loss or neurological disorders to express basic needs.
4. Human–Machine Interaction: Can be used in research and automation projects involving gesture-based control.
5. Rehabilitation Centers: Assists therapists and patients in communication and recovery sessions.

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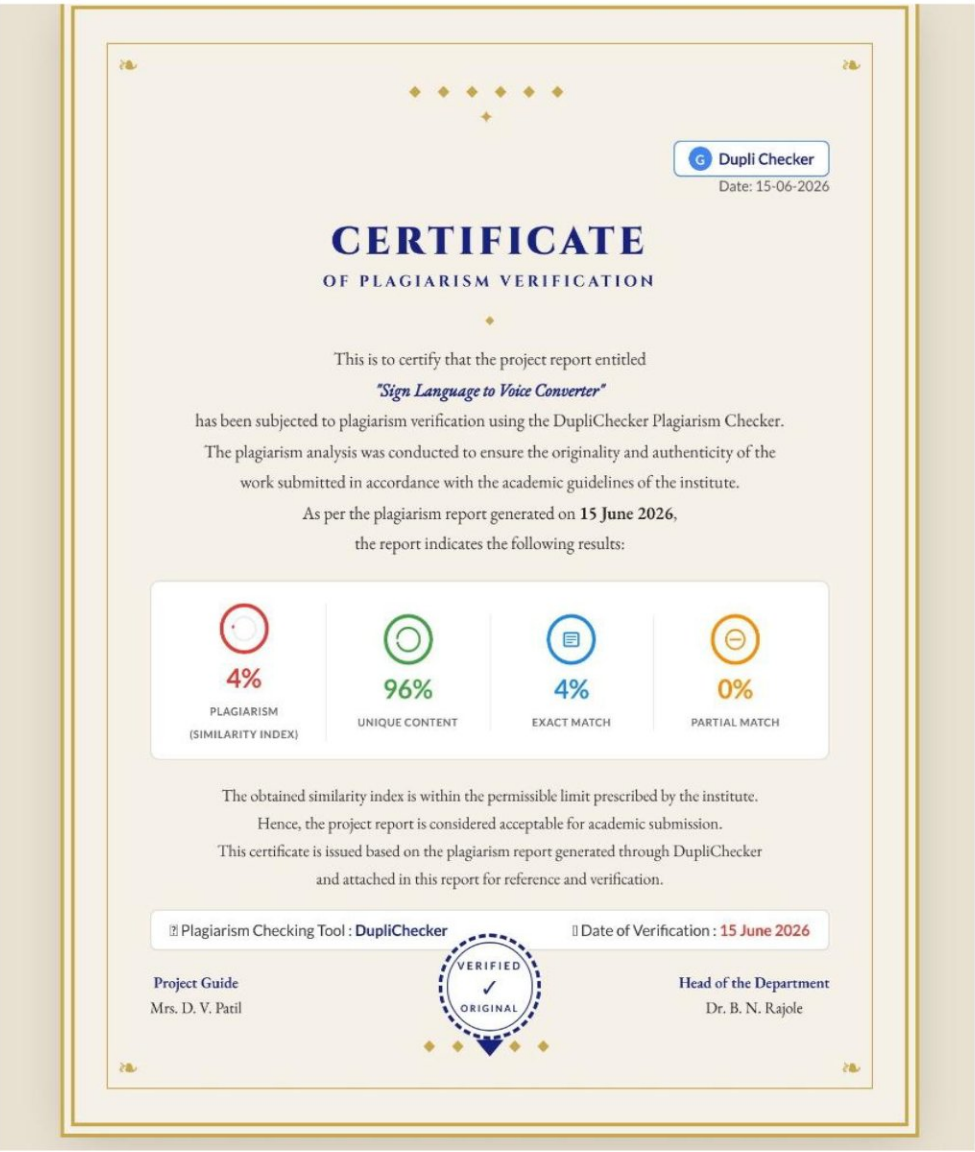

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